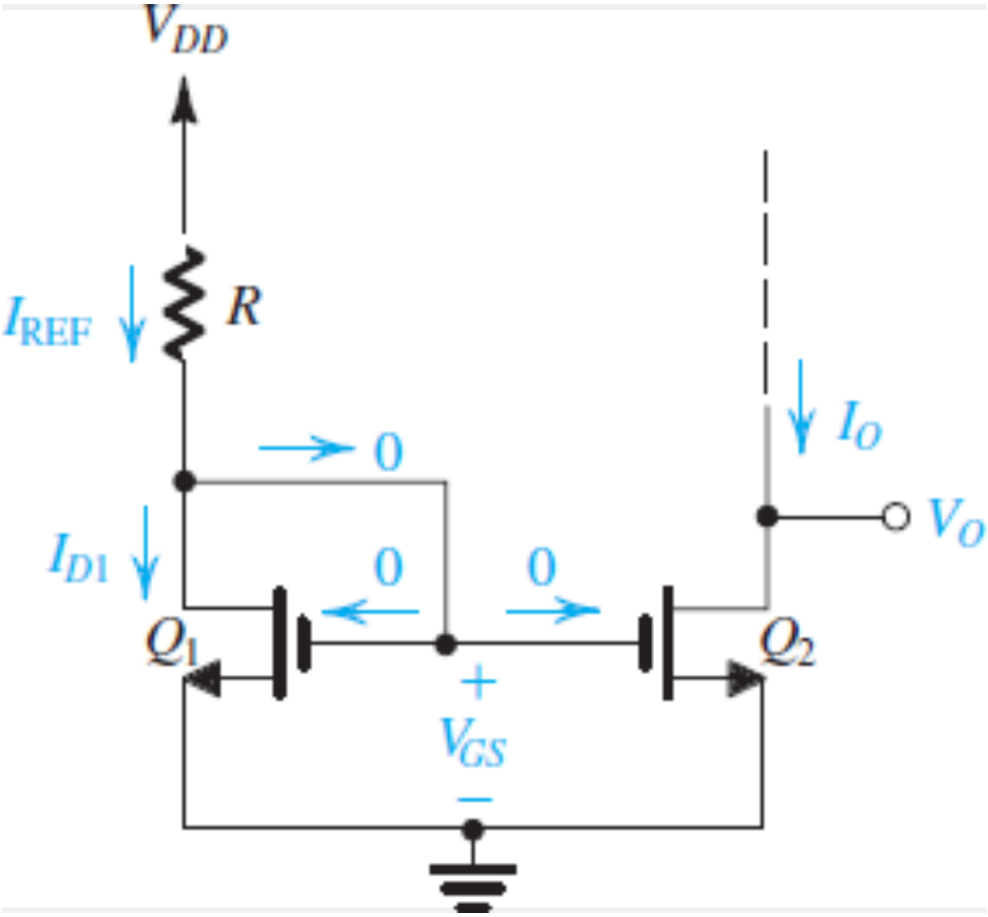
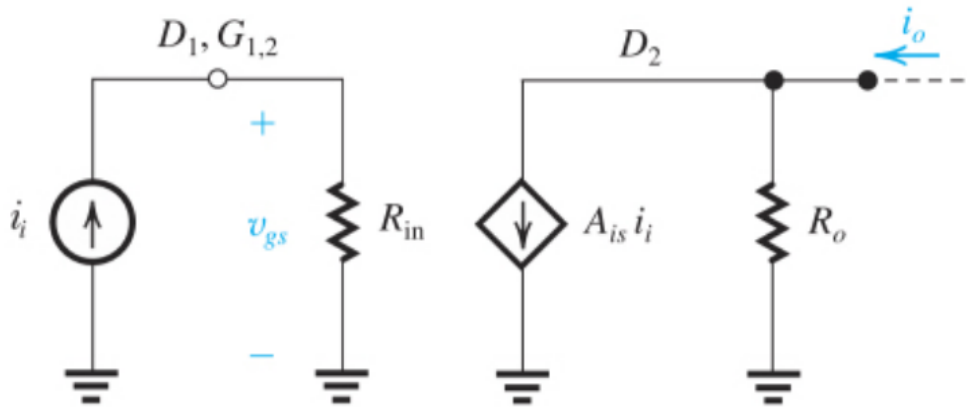
MOSFET Current Source

Schematic



Small-Signal Model



(b)

Equations

Equation	Conditions	Notes
$I_{D1} = I_{REF} = \frac{V_{DD} - V_{GS}}{R}$		Gate currents are zero
$I_O = I_{D2} = \frac{1}{2} k'_n \left(\frac{W}{L}\right)_2 (V_{GS} - V_t)^2$	$V_O \geq V_{GS} - V_t$	Must operate in satur
$I_O = \frac{(W/L)_2}{(W/L)_1} I_{REF} \left[1 + \frac{V_O - V_{GS}}{V_{A2}}\right]$		
$\Delta I_O = \frac{\Delta V_O}{r_{o2}} = \frac{V_O - V_{GS}}{r_{o2}}$		
$\frac{I_O}{I_{REF}} = \frac{(W/L)_2}{(W/L)_1}$	Negligible channel-length modulation	
$I_O = I_{REF}$	Identical transistors	
$A_{is} = \frac{g_{m2}}{g_{m1}} \frac{1}{1 + (1/g_{m1} r_{o1})}$		
$A_{is} = \frac{(W/L)_2}{(W/L)_1} \left[\frac{1}{1 + (1/g_{m1} r_{o1})}\right]$		

Notes

Term	Description
A_{is}	Small signal short-circuit current gain